



**UNIVERSITY OF EASTERN AFRICA, BARATON
SCHOOL OF BUSINESS**

DEPARTMENT OF INFORMATION, SYSTEMS, AND COMPUTING

FINAL PROJECT PROPOSAL

SENIOR PROJECT INSY 492

**TITLE: MUSIC TALENT TRACKING AND LOCATION SYSTEM (WAFUNGI-
NATION).**

BY

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PROJECT OVERVIEW

Wafungi-Nation project is a system that aims to create a comprehensive platform that connects musicians, event organizers, and instrument owners in a seamless and efficient manner. The system will allow musicians to showcase their skills, instruments, and availability, while enabling users to search, locate, and hire musicians based on specific criteria such as genre, instrument proficiency, location, and budget. Additionally, the platform will feature an instrument rental system, allowing users to rent musical instruments for personal or professional use. With integrated features like event posting, real-time notifications, payment processing, and reviews, the platform will serve as a one-stop solution for all music-related needs. By fostering collaboration and simplifying the process of finding talent or resources, this project seeks to empower the music community and create new opportunities for musicians.

BACKGROUND INFORMATION

The music industry thrives on collaboration, yet musicians, event organizers, and instrument owners often face challenges in connecting efficiently. Traditional methods of talent scouting such as word-of-mouth referrals, social media posts, and physical auditions are time-consuming and lack structured search capabilities. Similarly, instrument rentals are typically handled through niche stores or informal agreements, making it difficult for musicians to find affordable and reliable equipment. WAFUNGI-NATION aims to fill this gap by providing a **centralized digital marketplace** where musicians can showcase their skills, event organizers can find the right talent, and instrument owners can monetize their unused gear.

PROBLEM STATEMENT

Event organizers often struggle to locate skilled musicians who match their requirements in terms of genre, budget, and availability. Similarly, musicians face difficulties in showcasing their talents and connecting with potential opportunities. Additionally, instrument owners lack a centralized platform to rent out their equipment, while those in need of instruments often find it hard to locate affordable and reliable rentals. These inefficiencies lead to missed opportunities, wasted time, and limited growth for all parties involved. This project aims to address these challenges by developing a platform that streamlines the process of finding, hiring, and renting musical talent and resources, ultimately fostering a more connected and efficient music community.

PROBLEM SOLUTION

This proposed platform addresses the challenges faced by musicians, event organizers, and instrument owners by providing a centralized, user-friendly system. Musicians can create detailed profiles showcasing their skills, availability, and samples, making it easier for event organizers to find and hire the right talent. Event organizers can post their requirements, such as genre, budget, and location, and receive tailored recommendations or applications from qualified musicians. Additionally, the platform includes an instrument rental system, allowing instrument owners to list their equipment for rent and users to easily search, book, and pay for rentals. With features like real-time notifications, secure payments, and review systems, the platform streamlines collaboration, reduces inefficiencies, and creates new opportunities for all users. This solution bridges gaps in the music community, fostering connections and growth for musicians, organizers, and instrument owners alike.

RELEVANCE OF THE PROJECT

The Wafungi-Nation platform is relevant in today's music industry, where digital transformation is essential for growth and collaboration. The project will:

- Create job opportunities for musicians by enhancing their visibility.
- Facilitate efficient talent discovery for event organizers.
- Provide a structured marketplace for renting musical instruments.
- Foster collaboration and networking in the music community.

JUSTIFICATION

NEED FOR THE PROJECT

- **Musicians Struggle with Visibility:** Many talented artists lack platforms to effectively promote themselves to potential employers.
- **Event Organizers Waste Time Searching:** Finding musicians who match specific criteria (genre, budget, location) is often inefficient.
- **Instrument Rental Market is Fragmented:** No dedicated platform exists for renting musical equipment securely.

Benefits of WAFUNGI-NATION

For Musicians:

- Increased exposure to gig opportunities.
- Ability to set their rates and availability.
- Secure payment processing.

For Event Organizers:

- Quick access to vetted musicians.
- Filtering by genre, budget, and location.
- Reliable booking & contract management.

For Instrument Owners:

- Additional income by renting unused instruments.
- Controlled rental terms and secure transactions.

For the Music Industry:

- Reduced friction in talent hiring.
- Encourages collaboration and economic growth.

OBJECTIVES

1. To Develop a Centralized Platform for Musicians and Event Organizers: Create a user-friendly system that allows musicians to showcase their skills and availability, while enabling event organizers to search, locate, and hire musicians based on specific criteria such as genre, instrument proficiency, and location.

2. To Facilitate Instrument Rentals:

Build a feature that allows instrument owners to list their equipment for rent and enables users to easily search, book, and rent instruments for personal or professional use, complete with secure payment and scheduling options.

3. To Enhance Collaboration and Opportunities in the Music Community: Foster a connected and efficient ecosystem by providing tools for communication, reviews, and real-time notifications, empowering musicians, event organizers, and instrument owners to collaborate effectively and grow their networks.

SCOPE

- **Target Users:** Musicians, event planners, instrument owners, and renters.
- **Platform Features:**
 - User registration and profile management.
 - Search and filtering tools.
 - Booking and payment systems.
 - Admin dashboard for moderation.

FEASIBILITY ANALYSIS

Technical Feasibility:

The project will be developed using Django for the backend, an HTML-based frontend, and cloud-based hosting services to ensure scalability.

Operational Feasibility:

The system will require minimal training for users, making adoption easy for musicians and event organizers.

Economic Feasibility:

Open-source technologies reduce development costs.

The use of free-tier services for hosting, database, minimize initial investment.

The project can be monetized through commission on bookings, premium subscriptions, and advertising.

Legal Feasibility:

Compliance with digital transaction regulations and data privacy laws will be ensured to protect user information.

Operational Feasibility:

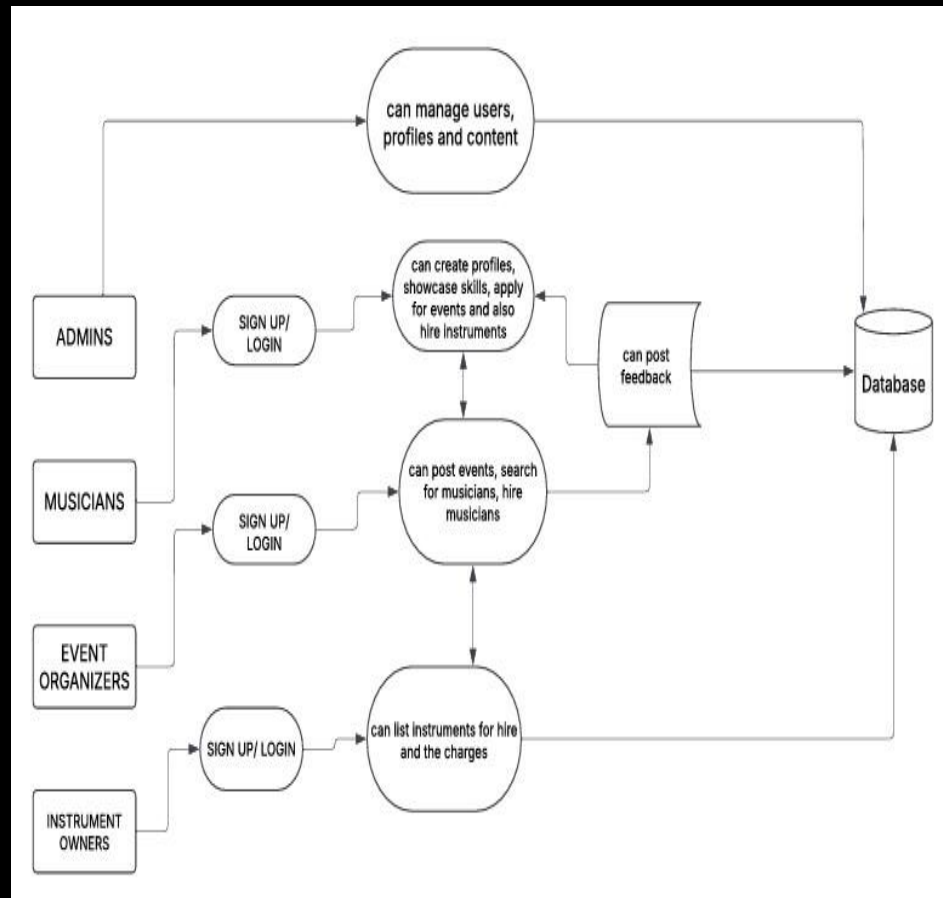
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TECHNOLOGIES USED

- ❖ Framework: Django (python-based web framework)
- ❖ Frontend: HTML, CSS, Javascript
- ❖ Backend: Django(python)

- ❖ Database: Sqlite
- ❖ APIs and Web Services: RESTful APIs for communication between frontend and backend
- ❖ Real-time notifications: Django Channels
- ❖ Development Environment: Visual Studio Code.
- ❖ Deployment tool: Heroku. A platform-as-a-service (PaaS)

CONCEPTUAL DIAGRAM



PROJECT TIMELINE

TASK	DURATION	DELIVERABLES	MILESTONES
Proposal planning	Week 1	The initial planning document	Outline and problem definition
Proposal writing	Week 1-2	Completing proposal document	Proposal document ready for review
Proposal presentation and approval	End of week 2	Presentation slides and supervisor approval	Approved project proposal
Requirements gathering	Week 3-4	Requirements specification document	Requirements ready
Data analysis	Week 4	Data analysis report	Requirements sign-off
System design	Week 5	System design document; complete conceptual diagram	Final system design
System development	Week 5-8	Developed modules; backend; admin setup, user authentication and landing page	Prototype of the system available
Mid-term progress report and presentation	Week 6	Mid-project progress report and presentation slides	Mid-project milestone reached
Project implementation	Week 9	Deployed prototype to be tested	Prototype deployed for testing
Testing	Week 10-11	Test reports	Confirmation of system stability
Project report documentation	Week 11-12	Final project report; documentation of code and modules	Complete project documentation
Presentation	Week 12	Final presentation slides; live demo	Successful final project presentation

BUDGET PROPOSAL

ITEM	QUANTITY	ITEM DESCRIPTION	COST ESTIMATION
Laptop	1	2.80 GHz processor 8GB Ram	28, 000
Internet	1	5 MBps net	5 000
Browser	1	Microsoft edge	free
Vs Code	1	Code editor	Free
Deployment and hosting	1		Free
Printing	1		500
TOTAL			Ksh 33,500

